‘Hack the University’ Challenge

ClassCompassAI

TIMG5001 – Assignment 2

Presented by: Katelyn Butts & Bhabna Mukherjee

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Problem Statement



There is no unified, predictive advising infrastructure that supports informed course planning at scale.

Students struggle to choose the right courses, while advisors juggle repetitive administrative tasks and disconnected systems.



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Stakeholders



Students

- Overwhelmed by so many options
- Uncertain about goal alignment
- Fear of making "wrong" choices leading to delays



Advisors

- Managing hundreds of students each term
- Manual prerequisite verification
- Lack of real-time risk visibility for student progress



Institution

- Delayed graduations impacting funding metrics
- Enrollment bottlenecks in popular courses
- Rising operational workload for advising departments

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Stakeholder Insights



Student Narrative

“There are too many courses; I don’t know what fits my goals.”

“Long descriptions and conflicting schedules overwhelm me.”



Advisors Narrative

“We spend hours checking prerequisites manually.”

“We can’t identify who is at academic risk early enough.”



Case Study insights

Students feel “digitally homeless” due to fragmented systems.

Faculty fear losing autonomy.

IT warns about integration cost & complexity.

How Might We Transform Academic Advising?

For Students

By providing **personalized & data-driven** course recommendations that reduce confusion and keep students on track toward their academic and career goals.

For Advisors

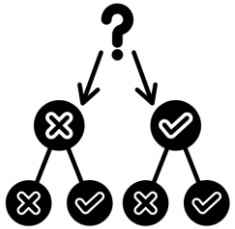
By **streamlining** routine administrative tasks and **seamlessly integrating information**, enabling advisors to shift their focus to proactive, high-value, and empathetic student support.

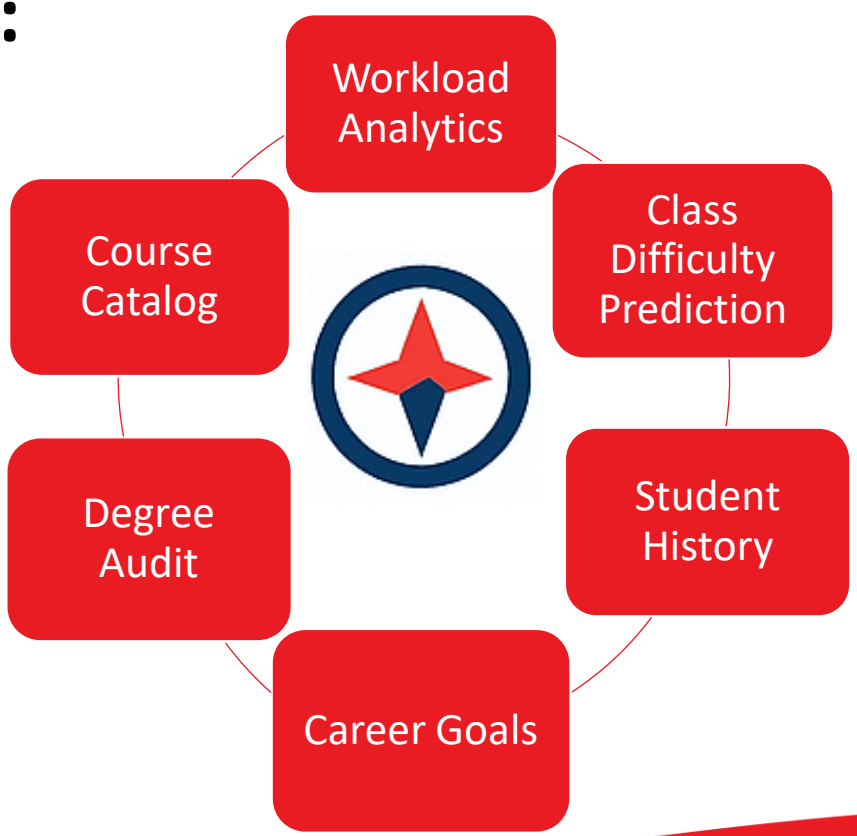
The Solution

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- Personalized Recommendations
- Visual Pathway Mapping
- Career-to-Course Skill Mapping
- Proactive Risk Alerts
- Integrated Advisor Dashboard
- 24/7 AI Course Guidance

How It Works:
System Overview





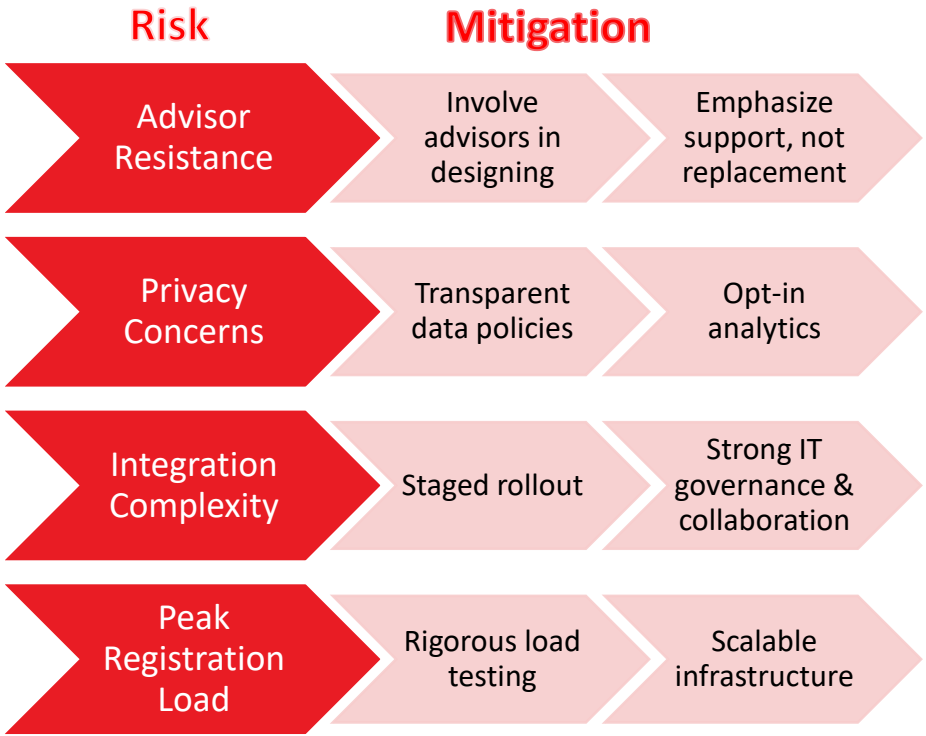
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Why Our Solution Wins

Criterion	Importance	How Course Compass Performs
Student Experience	High	Eliminates confusion; personalized guidance
Faculty Autonomy	High	Advisors control final decisions; tool supports, not replaces
Technical Feasibility	High	Uses existing systems (Banner, Degree Audit, Brightspace APIs)
Cost & Sustainability	Medium-High	Modular rollout; minimal initial infrastructure
Privacy & Security	High	Data minimization, opt-in analytics
Equity & Accessibility	High	Reduces advising inequities (international, first-gen students)
Strategic Alignment	High	Supports unified digital ecosystem vision from case study

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Risks Mitigation



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Major Cost Estimates

Machine Learning Assistance:	$300 – $800 CAD/ month
Development Hours:	300 – 400 total (internal or contract)
Training Programs:	$1,500 – $3,000 CAD
Ongoing Support:	$500 – $1,000 CAD/ month

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Execution Plan: A Phased Approach to Success

Phase 1 — Pilot Program

Focus: One Department (e.g., TIM/MABA)

- Conduct interviews and rule mapping
- Build Minimum Viable Product (MVP)
- Facilitate student and advisor testing
- Validate recommendations and risk alerts



Phase 2 — University-Wide Implementation

Focus: Expanding Reach & Integration

- Integrate Machine Learning models (difficulty/workload)
- Seamless integration with Brightspace & Carleton Central
- Comprehensive advisor training sessions

Phase 3 — Scale & Sustain

Focus: Continuous Improvement & Growth

- Rollout to all academic faculties
- Continuous model refinement and improvement
- Explore licensing opportunities with other Institutions

Future Vision

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For Students

- Smarter, more confident course choices
- Reduced academic stress and fewer delays
- Stronger alignment with career aspirations

For Advisors

- 30–40% fewer repetitive tasks
- More time for high-value, complex cases
- Enhanced capacity for proactive mentorship

For Carleton University

- Increased student retention rates
- Faster graduations, optimizing seat capacity
- Stronger institutional brand positioning
- New licensable technology asset







> Carleton ‘ClassCompass AI’ transforms academic advising from **reactive** to **predictive** — creating a smarter, more connected, and highly efficient university experience for everyone.

Thank You

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